True or false? (unless otherwise specified.)

Answer grid

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	B
True	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
False	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

1. **Degree of precision.** The degree of precision of the following quadrature rule is 2:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx \frac{1}{2}f(-1) + f(0) + \frac{1}{2}f(1)\,.$$

2. **Degree of precision.** The degree of precision of the following quadrature rule is 1:

$$\int_{0}^{1} f(x)\,\mathrm{d}x \approx f(0) + \frac{1}{2}f'(0)\,.$$

3. **Degree of precision.** Let $x_1 < \cdots < x_n$ be distinct nodes in $[-1, 1]$. If the quadrature rule

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx \sum_{i=1}^{n} w_i f(x_i)$$

has degree of precision at least $n - 1$, then its weights are given by

$$w_i = \int_{-1}^{1} \ell_i(x)\,\mathrm{d}x, \qquad \ell_i(x) = \prod_{\substack{j=1 \\ j\neq i}}^{n} \frac{x - x_j}{x_i - x_j}, \qquad i = 1, \ldots, n.$$

4. **Gaussian integration.** The following quadrature rule is the two-point Gauss–Legendre rule:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right). \tag{1}$$

5. **Composite quadrature.** Suppose that $f \in C^\infty[a, b]$ and let $I_n[f]$ denote the approximation of the integral $I[f] \coloneqq \int_a^b f(x)\,\mathrm{d}x$ obtained with the composite two-point Gauss–Legendre rule with n cells, that is

$$I_n[f] = \frac{h}{2}\sum_{i=0}^{n-1}\left(f\left(m_i - \frac{h}{2\sqrt{3}}\right) + f\left(m_i + \frac{h}{2\sqrt{3}}\right)\right), \qquad h = \frac{b-a}{n}, \qquad m_i = a + \left(i + \frac{1}{2}\right)h\,.$$

Then there exists a constant $C \in (0, \infty)$ such that

$$\forall n \in \{1, 2, \ldots\}, \qquad \Big|I[f] - I_n[f]\Big| \leqslant \frac{C}{n^4}\,.$$

Hint: Compute or recall the degree of precision of the rule (1), and use the theorem relating the degree of precision of a quadrature rule to the order of convergence, with respect to h, of the associated composite rule.

6. **Composite quadrature.** With the notation of the previous item, it holds that $I_n[f] = I[f]$ for all $n \geqslant 1$ whenever f is a polynomial of degree at most 3.

7. **Composite quadrature.** Assume that the error of a composite quadrature rule with n cells behaves like Cn^{-2} for large n, and that the error obtained with $n = 100$ cells is approximately 10^{-2}. Then the error obtained with $n = 1000$ cells is approximately 10^{-6}.

8. **Gaussian integration.** The one-point Gauss–Legendre rule on $[-1, 1]$ is the midpoint rule

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx 2f(0)\,.$$

9. **Gaussian integration.** The nodes of the n-point Gauss–Legendre rule on $[-1, 1]$ are equally spaced.

10. **Gaussian integration.** As for closed Newton–Cotes rules, the n-point Gauss–Legendre rule has some negative weights when n is large enough.

11. **Gaussian integration.** Let $x_1, \ldots, x_n$ and $w_1, \ldots, w_n$ denote the nodes and weights of the n-point Gauss–Legendre rule on $[-1, 1]$. Then the n-point Gauss–Legendre rule on a general interval $[a, b]$ is given by

$$\int_a^b f(x)\,\mathrm{d}x \approx \sum_{i=1}^{n} w_i f\left(a + \frac{b-a}{2}(x_i + 1)\right).$$

12. **Gaussian integration.** There exist a node $x_2 \in (-1, 1]$ and weights w_1, w_2 such that the rule

$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx w_1 u(-1) + w_2 u(x_2)$$

has a degree of precision equal to 2.

13. **Richardson extrapolation.** Let $T(h)$ be the composite trapezoidal approximation of $I := \int_a^b f(x)\,\mathrm{d}x$ with step size h, and assume that $T(h) = I + c_2 h^2 + c_4 h^4 + O(h^6)$ with $c_2 \neq 0$. Then the approximation

$$R(h) := 2T(h) - T(2h)$$

satisfies

$$\big|R(h) - I\big| = O(h^4).$$

14. **Richardson extrapolation.** With the notation and assumptions of question 13, it holds that

$$I - T(h) = \frac{T(h) - T(2h)}{3} + O(h^4).$$

In other words, the error of $T(h)$ can be estimated from $T(h)$ and $T(2h)$.

15. **Julia.** The following code implements the composite midpoint rule with n points:

```
function M(f, a, b, n)
    h = (b - a) / n
    x = LinRange(a + h/2, b - h/2, n + 1)
    return h * sum(f, x)
end
```

16. **Bonus.** Let $f \in C^\infty(\mathbb{R})$ and $x \in \mathbb{R}$, and let $D(h)$ denote the central finite difference approximation of $f'(x)$:

$$D(h) = \frac{f(x+h) - f(x-h)}{2h}.$$

Then the approximation

$$\frac{4D(h) - D(2h)}{3}$$

satisfies

$$\left|\frac{4D(h) - D(2h)}{3} - f'(x)\right| = O(h^4) \qquad \text{as } h \to 0.$$

Justify briefly.